

SAMPLE UNIVERSITY - END TERM EXAMINATION

Subject: Engineering Mathematics-I

Subject Code: BTAM-101

Date: 12 December 2025 Time: 3 Hours Maximum Marks: 60

Instructions: Attempt all sections. Use of calculator is not allowed.

Section-A (2 marks each)

Q1. Define a matrix and give one example.

Q2. State Rolle's theorem.

Section-B (5 marks each)

Q3. Find the rank of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Q4. Expand e^x using Maclaurin series.

Section-C (10 marks each)

Q5. Discuss the convergence of the series $\sum 1/n^2$.

Q6. Evaluate the double integral of xy over the unit square.